

or $(A\Psi, H\Psi) = \overline{(A\Psi, H\Psi)}$, for all $A \in \mathfrak{B}(\mathcal{H})_{\mathbb{R}}$. Hence $(\Psi, H\Phi) \in \mathbb{R}$ for all $\Phi \in \Psi^{\perp}$, which is possible only if $(\Psi, H\Phi)$ vanishes for all $\Phi \in \Psi^{\perp}$. This implies that Ψ must be an eigenvector of H . $\blacksquare$

From the symplectic point of view, the two steps in the construction of $\mathbb{P}\mathcal{H}$ appear in reverse order. Firstly, one pulls the symplectic form ω on $\mathcal{H}$ back to $\mathcal{S}\mathcal{H}$; here it is degenerate. Secondly, this degeneracy is removed upon quotienting $\mathcal{S}\mathcal{H}$ by $U(1)$, arriving at $\mathbb{P}\mathcal{H}$ once more. See IV.1.5.

2.6 Representations of Poisson Algebras

We look at symplectic manifolds as the classical analogues of modules for Poisson algebras (cf. the opening remark in 1.5).

Definition 2.6.1. A representation of a Poisson algebra $(\mathfrak{A}_{\mathbb{R}}, \circ, \{, \})$ is a linear map $\pi : \mathfrak{A}_{\mathbb{R}} \rightarrow C^{\infty}(S, \mathbb{R})$, where S is a symplectic manifold, satisfying

$$\begin{aligned} \pi(f \circ g) &= \pi(f)\pi(g); \\ \{\pi(f), \pi(g)\}_S &= \pi(\{f, g\}) \end{aligned} \quad (2.48)$$

(where $\{, \}_S$ is the Poisson bracket on S), as well as preserving completeness.

The condition (2.48) says simply that $\pi : \mathfrak{A}_{\mathbb{R}} \rightarrow C^{\infty}(S, \mathbb{R})$ is a morphism, assuming that the Jordan product in $C^{\infty}(S, \mathbb{R})$ is represented by pointwise multiplication (cf. 1.1.3). The completeness requirement means that the flow of $\xi_{\pi(h)}$ is defined for all times if h is complete in $\mathfrak{A}_{\mathbb{R}}$, cf. 2.3.4. It is imposed to eliminate constructions of the type $\mathfrak{A}_{\mathbb{R}} = C^{\infty}(P, \mathbb{R})$, $P' \neq P$ open in P , and π being simply restriction to P' .

There is a natural notion of equivalence. Namely, two representations $\pi_1 : \mathfrak{A}_{\mathbb{R}} \rightarrow C^{\infty}(S_1, \mathbb{R})$ and $\pi_2 : \mathfrak{A}_{\mathbb{R}} \rightarrow C^{\infty}(S_2, \mathbb{R})$ are called **equivalent** if there exists a symplectomorphism $J : S_1 \rightarrow S_2$ such that $J^*\pi_2(f) = \pi_1(f)$ for all $f \in \mathfrak{A}_{\mathbb{R}}$.

We can analyze the structure of representations of Poisson algebras of a slightly more general type than $C^{\infty}(P, \mathbb{R})$, where P is a Poisson manifold.

Definition 2.6.2. A Poisson space P is a Hausdorff topological space together with a linear subspace $\mathfrak{A}_{\mathbb{R}} \subset C(P, \mathbb{R})$ and a collection S_{α} of symplectic manifolds (called the symplectic leaves of P), as well as continuous injections $\iota_{\alpha} : S_{\alpha} \hookrightarrow P$, such that:

1. $P = \cup_{\alpha} \iota_{\alpha}(S_{\alpha})$ (disjoint union).
2. $\mathfrak{A}_{\mathbb{R}}$ separates points.
3. $\mathfrak{A}_{\mathbb{R}} \subseteq C_L^{\infty}(P, \mathbb{R})$, where $C_L^{\infty}(P, \mathbb{R})$ consists of all $f \in C(P, \mathbb{R})$ for which $\iota_{\alpha}^* f \in C^{\infty}(S_{\alpha}, \mathbb{R})$ for each α .
4. $\mathfrak{A}_{\mathbb{R}}$ is closed under the Poisson bracket

$$\{f, g\}(\iota_{\alpha}(\sigma)) = \{\iota_{\alpha}^* f, \iota_{\alpha}^* g\}_{\alpha}(\sigma). \quad (2.49)$$

If the ambient space P carries additional structure, such as a uniformity or a smooth structure, one can refine the above definition in the obvious way; such refinements will play an important role in later sections.

Definition 2.6.3. A **uniform Poisson space** is a Poisson space P in which the topology is defined by a uniformity on P and that satisfies Definition 2.6.2 with $C(P, \mathbb{R})$ replaced by the space $C_u(P, \mathbb{R})$ of uniformly continuous functions on P .

Similarly, a **smooth Poisson space** is a Poisson space for which P is a manifold and $C(P, \mathbb{R})$ is replaced by $C^\infty(P, \mathbb{R})$. By Theorem 2.4.7, a smooth Poisson space with $\mathfrak{A}_{\mathbb{R}} = C^\infty(P, \mathbb{R})$ is nothing but a Poisson manifold. The more general concept of a Poisson space is useful when the symplectic leaves do not fit together to form a manifold. This happens in the context of singular symplectic reduction, cf. IV.1.11. Moreover, we will show in 3.2.2 that the pure state space of a C^* -algebra is a uniform Poisson space. In any case, the object $C_L^\infty(P, \mathbb{R})$ defined in 2.6.2.3 is the function space intrinsically related to a (general, uniform, or smooth) Poisson space P .

Definition 2.6.2 does not entail that $\mathfrak{A}_{\mathbb{R}}$ is a Poisson algebra under pointwise multiplication as the Jordan product, but an interesting result arises when one makes that assumption. In preparation for this, we remark that the notion of a Poisson map makes sense in the context of Poisson spaces: It is still defined by (2.15).

Proposition 2.6.4. Let $(P, \mathfrak{A}_{\mathbb{R}})$ be a locally compact Poisson space for which $\mathfrak{A}_{\mathbb{R}}$ is a Poisson algebra under pointwise multiplication. If $\pi : \mathfrak{A}_{\mathbb{R}} \rightarrow C^\infty(S, \mathbb{R})$ is a representation of $\mathfrak{A}_{\mathbb{R}}$ on a finite-dimensional symplectic manifold S , then there exists a continuous map $J : S \rightarrow P$ such that $\pi = J^*$.

For simplicity we show this for compact P , and assume that $\mathfrak{A}_{\mathbb{R}}$ contains the unit function 1_P . The Stone–Weierstrass theorem then implies that $\mathfrak{A}_{\mathbb{R}}$ is dense in $C(P, \mathbb{R})$ in the sup-norm. Take a point $\sigma \in S$, and define a linear functional $\tilde{J}_\sigma$ on $\mathfrak{A}_{\mathbb{R}}$ by $\tilde{J}_\sigma(f) = (\pi(f))(\sigma)$. By the first member of (2.48), this functional is multiplicative. If it were defined on all of $C(P, \mathbb{R})$, we could immediately conclude from this that $\tilde{J}_\sigma$ is continuous; a positivity argument shows that this follows in the present case as well. Hence we extend $\tilde{J}_\sigma$ to all of $C(P, \mathbb{R})$. It follows that $\tilde{J}_\sigma$ defines a pure state, and pure states on $C(P, \mathbb{R})$ correspond to points of P (see 2.1.4). Hence $\tilde{J}_\sigma$ corresponds to a point $J(\sigma)$ in P , and this defines the desired map $J : S \rightarrow P$. The continuity of J follows from a technical argument in the theory of commutative C^* -algebras. The second member of (2.48) obviously implies that J is a Poisson map. $\square$

Corollary 2.6.5. If $\mathfrak{A}_{\mathbb{R}} = C^\infty(P, \mathbb{R})$ for a Poisson manifold P and $\pi : \mathfrak{A}_{\mathbb{R}} \rightarrow C^\infty(S, \mathbb{R})$ is a representation, then there exists a smooth Poisson map $J : S \rightarrow P$ such that $\pi = J^*$.

The smoothness of J follows from the property $\pi = J^*$. $\blacksquare$

There is a natural notion of irreducibility for representations of Poisson algebras.

Definition 2.6.6. A representation π of a Poisson algebra $\mathfrak{A}_{\mathbb{R}}$ is called **irreducible** if

$$\{\xi_{\pi(f)}(\sigma) \mid f \in \mathfrak{A}_{\mathbb{R}}\} = T_{\sigma} S \quad \forall \sigma \in S. \quad (2.50)$$

If S is infinite-dimensional, it is understood that one takes the closure of the left-hand side in the definition. The finite-dimensional irreducible representations of a Poisson algebra associated with a locally compact Poisson space (Definition 2.6.2) can be described concretely.

Theorem 2.6.7. Under the assumptions of Proposition 2.6.4, let π be irreducible. Then S is symplectomorphic to a symplectic leaf S_{α} of P , or to a covering space thereof.

In fact, it will follow from the proof below that an irreducible representation space S of a locally compact Poisson space has to be finite-dimensional. The proof of this theorem is based on 2.6.4; we have $\pi = J^*$ for $J : S \rightarrow P$. For each $\sigma \in S$, let $S_{\alpha(\sigma)}$ be the symplectic leaf for which $\iota_{\alpha(\sigma)}(S_{\alpha(\sigma)})$ contains $J(\sigma)$ (cf. 2.6.2); we will henceforth identify $\iota_{\alpha(\sigma)}(S_{\alpha(\sigma)})$ and $S_{\alpha(\sigma)}$. By irreducibility, any $X \in T_{\sigma} S$ can be written as $X = \xi_{\pi(f)}$ for some $f \in \mathfrak{A}_{\mathbb{R}}$. We define a linear map $J_* : T_{\sigma} S \rightarrow T_{J(\sigma)} S_{\alpha(\sigma)}$ by

$$J_* \xi_{\pi(f)}(\sigma) := \xi_f(J(\sigma)). \quad (2.51)$$

The notation is consistent: If P is a manifold, J_* is indeed the pushforward of J , cf. (2.17). The fact that J_* is well-defined follows from its injectivity, which we will now demonstrate. If $J_* \xi_{\pi(f)}(\sigma) = 0$, then $\{f, g\}(J(\sigma)) = 0$ for all $g \in \mathfrak{A}_{\mathbb{R}}$, since J is a Poisson map. But then $\omega_{\sigma}^S(\xi_{\pi(f)}, \xi_{\pi(g)}) = 0$ for all g , where ω^S is the symplectic form on S . Since ω^S is nondegenerate, this implies $\xi_{\pi(f)} = 0$, which proves injectivity. Now, J_* is evidently surjective as well, because $S_{\alpha(\sigma)}$ is symplectic. Hence J_* is an isomorphism.

Combining this result with Propositions 2.3.5 and 2.3.7, we conclude that $J(S) \subseteq S_{\alpha(\sigma)}$, where S and $S_{\alpha(\sigma)}$ are locally symplectomorphic (since π is a representation). The completeness of π (see Definition 2.6.1) implies that $J(S) = S_{\alpha(\sigma)}$. For if the inclusion $J(S) \subseteq S_{\alpha(\sigma)}$ were proper, we could take a neighborhood $\mathcal{N}$ of a boundary point of $J(S)$ in $S_{\alpha(\sigma)}$, and take $\sigma_1 \in \mathcal{N} \cap J(S)$ and $\sigma_2 \in \mathcal{N}$ but $\sigma_2 \notin J(S)$, such that σ_1 and σ_2 are connected by a Hamiltonian curve tangent to ξ_f (cf. 2.3.7). We then consider the Hamiltonian curve in S tangent to $\xi_{\pi(f)}$ and passing through s_1 , where $J(s_1) = \sigma_1$ (s_1 may not be unique). By 2.3.5 this curve is mapped onto the Hamiltonian curve connecting σ_1 and σ_2 , but this is impossible because of our assumptions on σ_2 . Hence the curve in S in question must suddenly stop somewhere, contradicting the completeness of π .

A similar argument shows that J is a covering projection. For J not to be a covering projection, there must exist a point $\sigma_3 \in S_{\alpha(\sigma)}$, a neighborhood $\mathcal{N}_{\sigma_3}$ of σ_3 , and a connected component $J_i^{-1}(\mathcal{N}_{\sigma_3})$ of $J^{-1}(\mathcal{N}_{\sigma_3})$ such that $J(J_i^{-1}(\mathcal{N}_{\sigma_3})) \subset \mathcal{N}_{\sigma_3}$ is a proper inclusion. But in that case we could choose points $\sigma_1 \in J(J_i^{-1}(\mathcal{N}_{\sigma_3}))$ and $\sigma_2 \in \mathcal{N}_{\sigma_3}$ but $\sigma_2 \notin J(J_i^{-1}(\mathcal{N}_{\sigma_3}))$, which can be connected by a smooth Hamiltonian curve, tangent to some vector field ξ_g . Let $\sigma_1 = J(s_1)$ for some s_1 , and consider the

flow of $\xi_{\pi(g)}$ through s_1 in S . Then this flow must either suddenly stop, contradicting the completeness of π , or continue outside $J_i^{-1}(\mathcal{N}_\sigma)$ to a point s_2 for which $J(s_2) = \sigma_2$, contradicting the assumptions on σ_2 . Hence J must be a covering projection, and Theorem 2.6.7 is proved. ■

We now return to C^* -algebras and their pure state spaces. Take a C^* -algebra $\mathfrak{A}$ with pure state space $\mathcal{P} = \mathcal{P}(\mathfrak{A})$ (equipped with the w^* -topology), and identify its self-adjoint part $\mathfrak{A}_\mathbb{R}$ with a subspace of $C(\mathcal{P}, \mathbb{R})$ by the Gelfand transform $\hat{A} \in C(\mathcal{P}, \mathbb{R})$; see (1.30). We will occasionally drop the hat on A .

Proposition 2.6.8. *The pure state space $\mathcal{P} = \mathcal{P}(\mathfrak{A})$ of a C^* -algebra $\mathfrak{A}$ (where $\mathfrak{A}_\mathbb{R}$ is identified with a subspace of $C(\mathcal{P}, \mathbb{R})$ through the Gelfand transform (1.30)), equipped with the irreducible representation spaces $S_\alpha = \mathbb{P}\mathcal{H}_\alpha$ and the inclusion maps ι_α , is a Poisson space.*

This is a trivial consequence of 2.5.4; note that the Poisson bracket in the sense of 2.6.2 coincides with the one (1.22) originally defined on $\mathfrak{A}_\mathbb{R}$. Recall that the choice of each $\mathcal{H}_\alpha$ is arbitrary within unitary equivalence; the Poisson structure on $\mathcal{P}(\mathfrak{A})$ is independent of the particular choices made by the comment following (2.42). The spaces $\mathbb{P}\mathcal{H}_\alpha$ are now seen to be the symplectic leaves of $\mathcal{P}$. ■

Proposition 2.6.8 recognizes the fact that (the self-adjoint parts of) C^* -algebras fall under the theory of Poisson spaces. This point of view receives further support from a reconsideration of the notion of a representation π of a C^* -algebra on a Hilbert space $\mathcal{H}$ (see 1.5.1). As explained in 2.5, we may identify $\mathfrak{B}(\mathcal{H})_\mathbb{R}$ with a subspace of the Poisson algebra $C^\infty(\mathcal{H}, \mathbb{R})$, so that π maps $A \in \mathfrak{A}_\mathbb{R}$ to $\widehat{\pi(A)} \in C^\infty(\mathbb{P}\mathcal{H}, \mathbb{R})$ (cf. (2.43)). It follows from 2.5.6 that $\pi : \mathfrak{A}_\mathbb{R} \rightarrow C^\infty(\mathcal{H}, \mathbb{R})$, thus interpreted, is a Poisson morphism.

Proposition 2.6.9. *A representation π of a C^* -algebra $\mathfrak{A}$ on a Hilbert space $\mathcal{H}$ is irreducible iff for every $\psi \in \mathbb{P}\mathcal{H}$ the set $\{v(i\pi(A)\psi) \mid A \in \mathfrak{A}_\mathbb{R}\}$ of tangent vectors is dense in $T_\psi\mathbb{P}\mathcal{H}$ (Poisson irreducibility).*

This follows from (2.31) and 2.2.2.2. Note that $T_\psi\mathcal{H}$ equals $\{V(A\psi) \mid \psi \in \mathfrak{B}(\mathcal{H})\}$, but does not equal $\{V(A\psi) \mid \psi \in \mathfrak{B}(\mathcal{H})_\mathbb{R}\}$; nonetheless, $T_\psi\mathbb{P}\mathcal{H}$ is given by (2.31). This is because the orthogonal complement of $\{V(A\psi) \mid \psi \in \mathfrak{B}(\mathcal{H})_\mathbb{R}\}$ in $T_\psi\mathcal{H}$ projects to zero in $T_\psi\mathbb{P}\mathcal{H}$. ■

Combining 2.6.9 and (2.45), we see that the notions of irreducibility of a representation of a C^* -algebra (Definition 2.2.1) and of a Poisson algebra (Definition 2.6.6) coincide (cf. 2.2.1). Therefore, on the Poisson side there is a close formal similarity between C^* -algebras and Poisson algebras as far as their respective representation theories are concerned. Indeed, combining Theorems 2.5.4 and 2.6.7 and Proposition 2.6.8, we obtain (under the above identifications)

Corollary 2.6.10. *Let $\mathfrak{A}_\mathbb{R}$ be either a Poisson algebra defined by a locally compact Poisson space P , or the self-adjoint part of a C^* -algebra with pure state space $\mathcal{P}$. Then, up to equivalence, every irreducible representation of $\mathfrak{A}_\mathbb{R}$ is given by the*

restriction of $\mathfrak{A}_{\mathbb{R}}$ to a symplectic leaf of P (or $\mathcal{P}$, respectively), or by this restriction preceded by the covering projection on a covering space of such a leaf.

Note that $\mathbb{P}\mathcal{H}$ has no nontrivial covering spaces. If the Poisson space P is compact and the Poisson algebra $\mathfrak{A}_{\mathbb{R}}$ contains the unit function, then $\mathfrak{A}_{\mathbb{R}}$ is dense in $C(P, \mathbb{R})$ (cf. the proof of 2.6.4); in that case, P is actually the pure state space of $\mathfrak{A}$ as a C^* -algebra (cf. 2.1). If P is merely locally compact, the same conclusion holds if $\mathfrak{A}_{\mathbb{R}}$ is contained in $C_0(P)$.

The difference between representations of C^* -algebras and Poisson algebras lies on the Jordan side; from the point of view of pure states, the Jordan structure on $\mathfrak{A}_{\mathbb{R}}$ eventually originates from a novel structure on $\mathcal{P}(\mathfrak{A})$.

2.7 Transition Probability Spaces

Here is the structure alluded to at the end of the preceding section.

Definition 2.7.1. A transition probability on a set $\mathcal{P}$ is a function

$$p : \mathcal{P} \times \mathcal{P} \rightarrow [0, 1] \quad (2.52)$$

that satisfies

$$p(\rho, \sigma) = 1 \iff \rho = \sigma \quad (2.53)$$

and

$$p(\rho, \sigma) = 0 \iff p(\sigma, \rho) = 0. \quad (2.54)$$

A set with such a transition probability is called a **transition probability space**.

The following set of definitions is natural and self-evident.

Definition 2.7.2. A family of subsets of a transition probability space $\mathcal{P}$ is called **orthogonal** if $p(\rho, \sigma) = 0$ whenever ρ and σ do not lie in the same subset. The space $\mathcal{P}$ is called **reducible** if it is the union of two (nonempty) orthogonal subsets; if not, it is said to be **irreducible**. A **component** C of $\mathcal{P}$ is a subset $C \subset \mathcal{P}$ such that C and $\mathcal{P} \setminus C$ are orthogonal. An irreducible component of $\mathcal{P}$ is called a **sector**.

Thus any transition probability space is the disjoint union of its sectors.

Certain subsets of $\mathcal{P}$ are of special significance. The **orthoplement** of $Q \subset \mathcal{P}$ is defined by

$$Q^\perp = \{\sigma \in \mathcal{P} \mid p(\rho, \sigma) = 0 \forall \rho \in Q\}. \quad (2.55)$$

It is immediately obvious that if $R \subseteq S$, then $S^\perp \subseteq R^\perp$, and that $T \subseteq T^{\perp\perp}$. Putting $R = Q$ and $S = Q^{\perp\perp}$ shows that $Q^{\perp\perp\perp} \subseteq Q^\perp$; putting $T = Q^\perp$ yields $Q^\perp \subseteq Q^{\perp\perp\perp}$. Hence $Q^\perp = Q^{\perp\perp\perp}$. Accordingly, the **orthoclosure** of a subset $Q \subseteq \mathcal{P}$ is defined as $Q^{\perp\perp}$, and Q is called **orthoclosed** if $Q = Q^{\perp\perp}$. It follows that $Q^\perp$ (and therefore $Q^{\perp\perp}$) is always orthoclosed. Also, one easily sees that any component $C \subseteq \mathcal{P}$ is orthoclosed; this applies in particular to $\mathcal{P}$ itself, and to any sector of $\mathcal{P}$.